

# Changyu (Thomason) Zhao

608-886-7470 | [czhao249@wisc.edu](mailto:czhao249@wisc.edu) | [linkedin.com/in/changyu-zhao](https://www.linkedin.com/in/changyu-zhao) | [github.com/ThomasonZhao](https://github.com/ThomasonZhao)

## EDUCATION

### University of Wisconsin - Madison

*Bachelor of Science in Computer Science (Honors) and Math*

Sep. 2021 – May 2025

*Madison, Wisconsin*

## EXPERIENCE

### UW - Madison Cybersecurity Operation Center

*Cybersecurity Student Intern Analyst*

Feb. 2023 – Present

*Madison, Wisconsin*

- Analyzed daily outbound malicious traffic reports from Palo Alto Firewall and Cisco Secure Endpoint (AMP).
- Traced malicious traffic in Elastic and Palo Alto Panorama to notify security alert to related user or department.
- Conducted security authorization and Identity Access Management (IAM) for access restrictive accounts.

### Tencent Spark Early Career

*Cybersecurity Researcher Intern*

Jun. 2021 – Present

*Shenzhen, China*

- Through source code review to find the deserialization and sandbox escape vulnerability of modified Vijos Online Judge (OJ) system. Successfully attacked by gaining the reverse root privileged shell under docker container.
- Escalated privilege to root by escaping from privileged docker container by abusing CVE-2019-5736.
- Utilized IDA to reverse engineering the router firmware and locate buffer overflow mentioned in CVE-2018-8879. Applied Returned Oriented Programming (ROP) skills to get reverse shell and fully control the router.

## PROJECTS

### Tenda AC18 AC1900 Router Exploitation | IDA, GDB, QEMU, Python

May 2022 – Aug.2022

- Exploited unfixed Common Vulnerabilities and Exposures (CVE) on Tenda AC18 router firmware.
- Applied qemu-system to emulate arm32 router firmware on x86-64 machine for systematic debugging.
- Developed decompiled code level automated stack overflow and command injection detection plug-in for IDA.
- Achieved Remote Code Execution (RCE) through command injection and buffer overflow vulnerability.
- Discovered unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.

### Windows Calculator Reverse Engineering | IDA, Windbg, Frida

Jan. 2022 – Apr. 2022

- Exploited Win10 & Win32 Calculator to return absolutely wrong arithmetic result, for example:  $1 + 1 = 3$ .
- Utilized IDA and Windbg to reverse engineering and do both static and dynamic analysis on the calculator binary.
- Utilized Frida to apply hooks on the calculator's library functions to hijack arbitrary arithmetic operation.

## EXTRACURRICULAR ACTIVITIES

### r3kapig CTF Team

*Team member*

Sep. 2021 – Present

- DEF CON CTF Finals, Rank 12/1282 world wide.
- Competed in more than 20 48-hour CTFs since 2021; Specialized in Binary Exploitation & Reverse Engineering

### pwn.college

*Independent Learning*

Dec. 2021 – Jul. 2022

*Tempe, Arizona*

- Earned yellow belt and blue belt (Finish all 472 challenges in the dojo instance). Current ranking 43/2500+.
- Learned binary exploitation skills: program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race condition, and kernel security.

## TECHNICAL SKILLS

**Languages:** C/C++, Java, Python, x86-Assembly JavaScript, HTML/CSS,  $\text{\LaTeX}$

**Binary Security Tools/Framework:** IDA/Ghidra, GDB/Windbg, Angr, Frida, Elastic Search, Joern

**Web Security Tools:** Burpsuit, Wireshark, Metasploit Framework, Kali Linux